

Contextualized Representation for Corpus-based Chatbots

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آکادمی‌ها



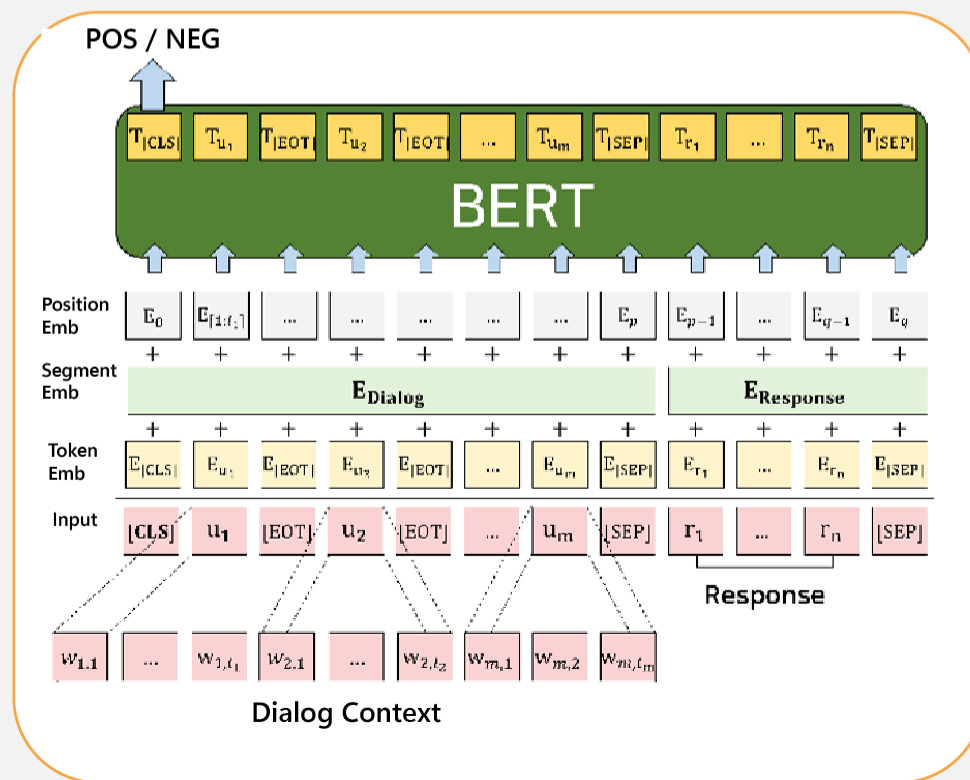
➤ **Corpus-based**

- ▶ **Retrieval Methods**
- ▶ **Generation Methods**



Retrieval Methods with BERT

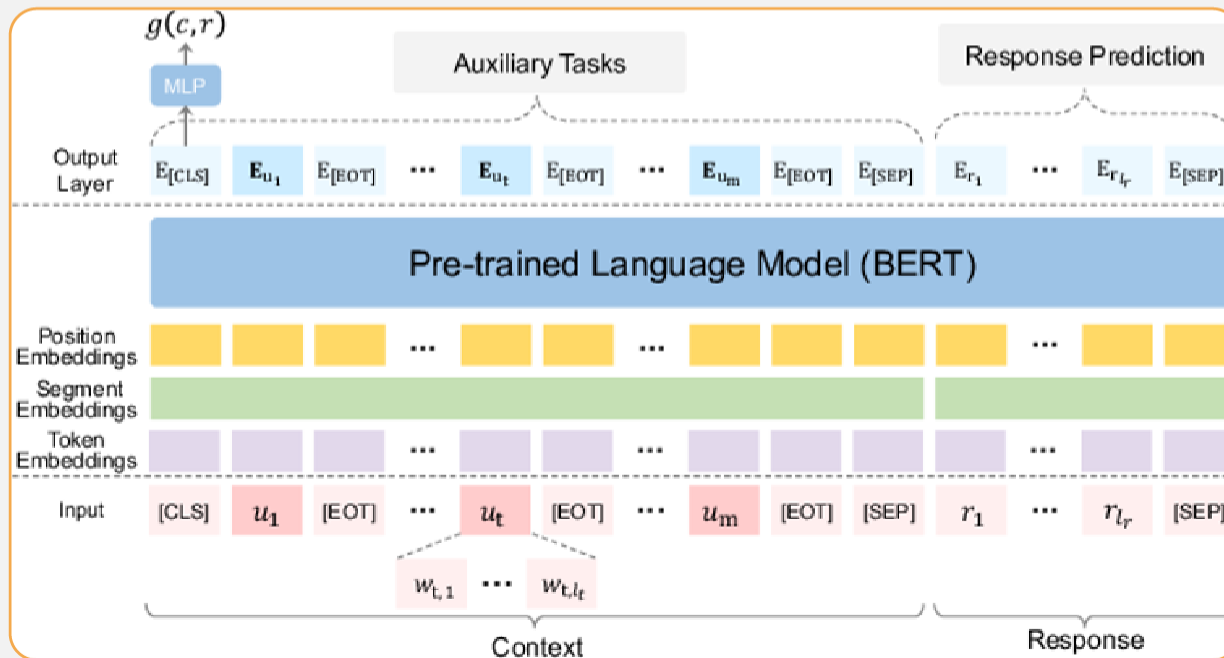
► An Effective Domain Adaptive Post-Training Method for BERT in Response Selection





Retrieval Methods with BERT

► Learning an Effective Context-Response Matching Model with Self-Supervised Tasks for Retrieval-based Dialogues





Retrieval Methods with BERT

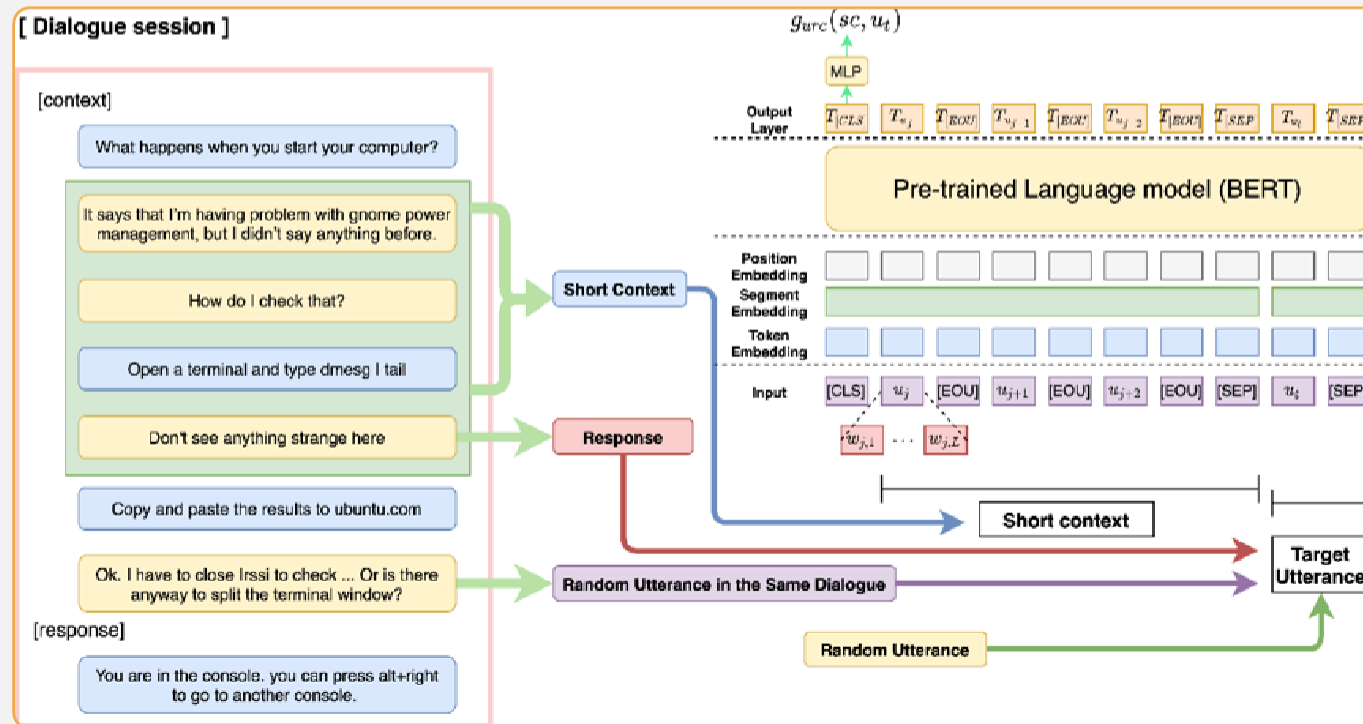
➤ Speaker-Aware BERT for Multi-Turn Response Selection in Retrieval-Based Chatbots

Input	[CLS]	How	are	you	[EOU]	[EOT]	Go	to	hold	a	drum	class	[EOU]	Anyone	joins	[EOU]	[EOT]	Sure	[EOU]	[EOT]	[SEP]	Have	fun	in	class	[SEP]
Token Embeddings	$E_{[CLS]}$	E_{How}	E_{are}	E_{you}	$E_{[EOU]}$	$E_{[EOT]}$	E_{Go}	E_{to}	E_{hold}	E_a	E_{drum}	E_{class}	$E_{[EOU]}$	E_{Anyone}	E_{joins}	$E_{[EOU]}$	$E_{[EOT]}$	E_{Sure}	$E_{[EOU]}$	$E_{[EOT]}$	$E_{[SEP]}$	E_{Have}	E_{fun}	E_{in}	E_{class}	$E_{[SEP]}$
	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Segment Embeddings	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_A	E_B	E_B	E_B	E_B	E_B	E_B
	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Position Embeddings	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7	E_8	E_9	E_{10}	E_{11}	E_{12}	E_{13}	E_{14}	E_{15}	E_{16}	E_{17}	E_{18}	E_{19}	E_{20}	E_{21}	E_{22}	E_{23}	E_{24}	E_{25}
	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+
Speaker Embeddings	E_0	E_0	E_0	E_0	E_0	E_0	E_1	E_1	E_1	E_1	E_1	E_1	E_1	E_1	E_1	E_1	E_1	E_0	E_0	E_0	E_0	E_1	E_1	E_1	E_1	E_1



Retrieval Methods with BERT

► Fine-grained Post-training for Improving Retrieval-based Dialogue Systems





► Fine-grained Post-training for Improving Retrieval-based Dialogue Systems

Models	Ubuntu			Douban						E-commerce		
	$R_{10}@1$	$R_{10}@2$	$R_{10}@5$	MAP	MRR	$P@1$	$R_{10}@1$	$R_{10}@2$	$R_{10}@5$	$R_{10}@1$	$R_{10}@2$	$R_{10}@5$
TF-IDF (Lowe et al., 2015)	0.410	0.545	0.708	0.331	0.359	0.180	0.096	0.172	0.405	0.159	0.256	0.477
RNN (Lowe et al., 2015)	0.403	0.547	0.819	0.390	0.422	0.208	0.118	0.223	0.589	0.325	0.463	0.775
CNN (Kadlec et al., 2015)	0.549	0.684	0.896	0.417	0.440	0.226	0.121	0.252	0.647	0.328	0.515	0.792
LSTM (Kadlec et al., 2015)	0.638	0.784	0.949	0.485	0.537	0.320	0.187	0.343	0.720	0.365	0.536	0.828
SMN (Wu et al., 2017)	0.726	0.847	0.961	0.529	0.569	0.397	0.233	0.396	0.724	0.453	0.654	0.886
DUA (Zhang et al., 2018)	0.752	0.868	0.962	0.551	0.599	0.421	0.243	0.421	0.780	0.501	0.700	0.921
DAM(Zhou et al., 2018)	0.767	0.874	0.969	0.550	0.601	0.427	0.254	0.410	0.757	0.526	0.727	0.933
IOI (Tao et al., 2019)	0.796	0.894	0.974	0.573	0.621	0.444	0.269	0.451	0.786	0.563	0.768	0.950
ESIM (Chen and Wang, 2019)	0.796	0.894	0.975	-	-	-	-	-	-	0.570	0.767	0.948
MSN (Yuan et al., 2019)	0.800	0.899	0.978	0.587	0.632	0.470	0.295	0.452	0.788	0.606	0.770	0.937
BERT (Gu et al., 2020)	0.808	0.897	0.975	0.591	0.633	0.454	0.280	0.470	0.828	0.610	0.814	0.973
RoBERTa-SS-DA (Lu et al., 2020)	0.826	0.909	0.978	0.602	0.646	0.460	0.280	0.495	0.847	0.627	0.835	0.980
BERT-DPT (Whang et al., 2020)	0.851	0.924	0.984	-	-	-	-	-	-	-	-	-
BERT-VFT (Whang et al., 2020)	0.855	0.928	0.985	-	-	-	-	-	-	-	-	-
SA-BERT (Gu et al., 2020)	0.855	0.928	0.983	0.619	0.659	0.496	0.313	0.481	0.847	0.704	0.879	0.985
UMS _{BERT+} (Whang et al., 2021)	0.875	0.942	0.988	0.625	0.664	0.499	0.318	0.482	0.858	0.762	0.905	0.986
BERT-SL (Xu et al., 2021)	0.884	0.946	0.990	-	-	-	-	-	-	0.776	0.919	0.991
BERT-FP (diff. %p)	0.911 (+2.7)	0.962 (+1.6)	0.994 (+0.4)	0.644 (+1.9)	0.680 (+1.6)	0.512 (+1.3)	0.324 (+0.6)	0.542 (+4.7)	0.870 (+1.2)	0.870 (+9.4)	0.956 (+3.7)	0.993 (+0.2)



» Corpus-based

- ▶ Retrieval Methods
- ▶ **Generation Methods**



Generation Methods with GPT

- An Empirical Investigation of Pre-Trained Transformer Language Models for Open-Domain Dialogue Generation

	Context Prediction				Response Prediction			
Output	x_1	x_2	x_3	$\langle \text{bos} \rangle$	y_0	y_1	y_2	$\langle \text{eos} \rangle$
Positional Embeddings	E_0	E_1	E_2	E_3	E_4	E_5	E_6	E_7
	+	+	+	+	+	+	+	+
Token Embeddings	E_{x_0}	E_{x_1}	E_{x_2}	E_{x_3}	$E_{\langle \text{bos} \rangle}$	E_{y_0}	E_{y_1}	E_{y_2}
Input	x_0	x_1	x_2	x_3	$\langle \text{bos} \rangle$	y_0	y_1	y_2



Word Frequency

DLGNet: A Transformer-based Model for Dialogue Response Generation



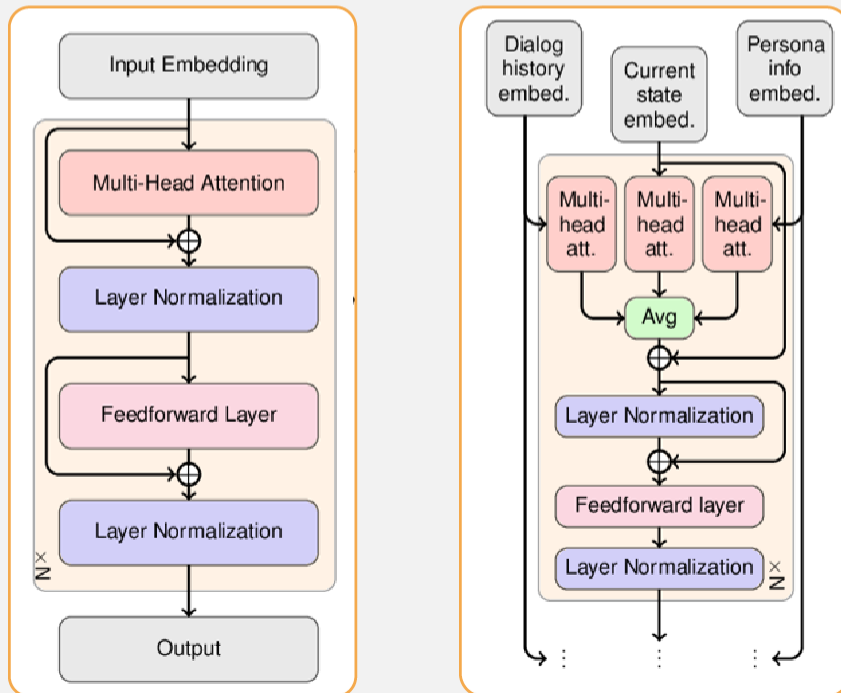
DLGNet input and output example consisting of a 3-turn conversation sample

- ▶ separated by [TSEP] token
- ▶ combined with random informative paddings, before and after
- ▶ Paddings and conversations are separated by [CSEP] tokens



Generation Methods with GPT

➤ Large-Scale Transfer Learning for Natural Language Generation





Further Reading

➤ Papers